Product Analytics Dashboard: User Engagement & Adoption Insights

GitHub: Project View

Introduction

This project focuses on analyzing user behavior and platform adoption for an analytics platform using Power BI. The main objective is to understand how users interact with different features, identify engagement patterns, and track overall usage of the platform.

A synthetic dataset was created to simulate realistic user activity, including feature usage and step-by-step progression through the platform. The dashboard highlights key metrics such as daily active users, session activity, and user journey through different stages.

The goal of this analysis is to provide clear insights into user behavior and identify areas where users drop off, which can help improve overall platform performance.

Data Model

The data model for this project is designed to represent user activity and platform interactions in a structured way. It consists of five main tables: Users, Events, Sessions, Features, and Date.

The Users table contains information about each user, including their signup date, country, and plan type. The Events table captures user actions such as login, data upload, and other feature usage. The Sessions table stores session-level information, including session start time and duration.

A Date table is used to enable time-based analysis and is connected to both Events and Sessions tables. The Users table is connected to Events and Sessions using user_id, allowing analysis of user behavior across different activities.

This structure ensures a clean and efficient data model, supporting accurate filtering and analysis across different dimensions.

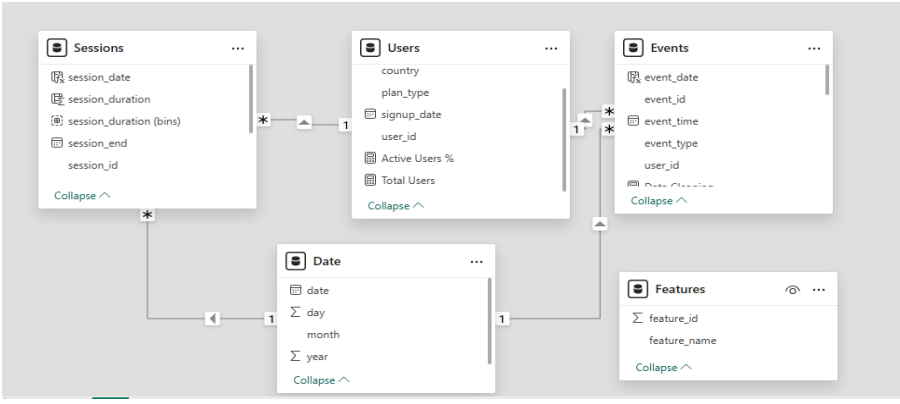

Figure: Data Model Diagram

Data Generation & Key Metrics

The dataset used in this project was generated manually with realistic patterns to simulate real-world user behavior on an analytics platform. Instead of using purely random data, logical patterns were introduced to reflect realistic usage.

For example, user activity was designed to follow a funnel flow (Login → Data Upload → Data Cleaning → EDA → Visualization → ML → Export), with a gradual drop-off at each stage. Additionally, differences

between Free and Pro users were incorporated, where Pro users show higher engagement in advanced features such as ML and Export.

To analyze platform performance, several key metrics were defined. These include Total Users, Daily Active Users (DAU), Total Sessions, and Active Users Percentage. Funnel-based metrics such as EDA % and Export % were used to measure user progression, along with Power Users to identify highly engaged users.

These metrics help in understanding both overall engagement and detailed user behavior across different stages of the platform.

Dashboard Overview

The Power BI dashboard is divided into two main sections to provide both a high-level overview and detailed user behavior analysis.

The first page, *Executive Overview*, focuses on overall platform performance. It includes key metrics such as Total Users, Daily Active Users (DAU), Total Sessions, Average Session Duration, and Active Users Percentage. Visualizations such as feature usage, DAU trend, user distribution by country, signup trends, and plan type distribution help in understanding overall engagement and growth.

Interactive slicers for date, country, and plan type are also included to allow dynamic filtering and deeper analysis.

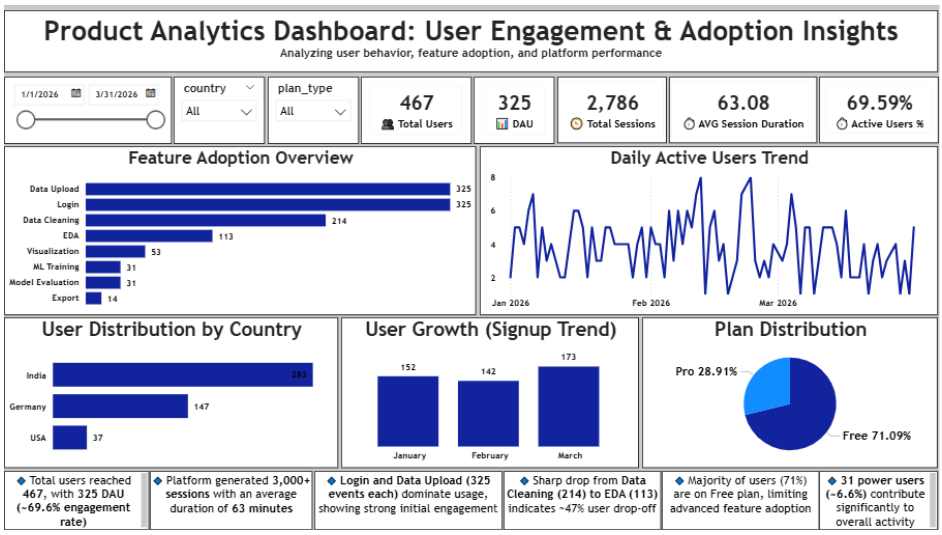

Figure: Executive Overview Dashboard

The second page, *User Journey & Behaviour Analysis*, provides a deeper view of how users interact with the platform. It includes a funnel analysis to track user progression across different stages, feature usage by plan type, and session activity trends. Additional visuals such as session duration distribution and power user identification help in analyzing user engagement in detail.

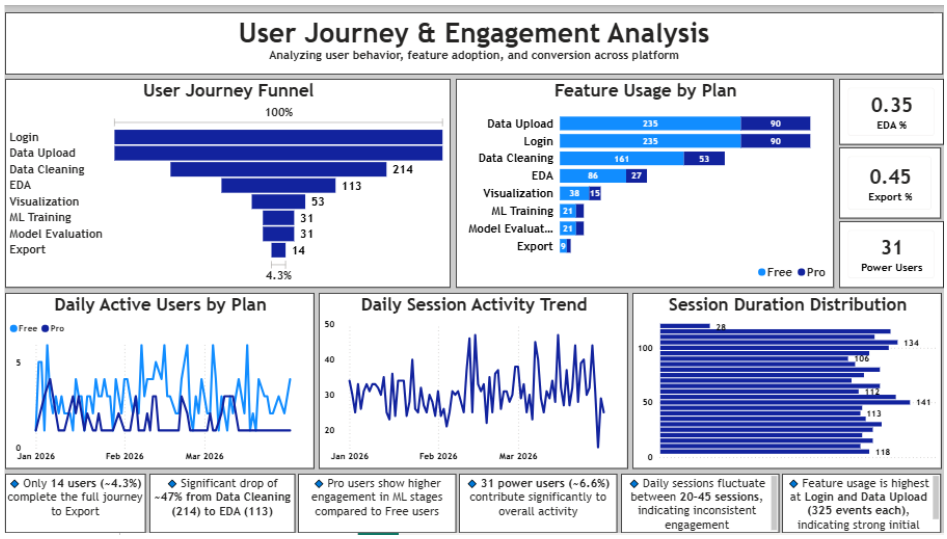

Figure: User Journey & Behaviour Analysis Dashboard

Together, these two pages provide a comprehensive view of both high-level metrics and detailed user behavior patterns.

Key Insights

The analysis of user behavior and platform usage reveals several important insights:

• The platform has a total of 467 users, with around 325 daily active users, resulting in an engagement rate of approximately 69.6%.

• User activity is highly concentrated in the initial stages, with Login and Data Upload each recording 325 events, indicating strong initial engagement.

• A significant drop-off of nearly 47% is observed between Data Cleaning (214 users) and EDA (113 users), highlighting a critical stage where users disengage.

• Only 14 users (~4.3%) complete the full journey up to Export, showing very low end-to-end conversion.

• Pro users demonstrate higher engagement in advanced features such as ML Training and Model Evaluation compared to Free users.

• A small group of 31 power users (~6.6%) contributes disproportionately to overall activity, indicating a highly engaged core user base.

Conclusion

The dashboard gives a clear view of how users are interacting with the platform and how different features are being used. It also highlights where users are dropping off during their journey.

While users are quite active in the beginning, there is a noticeable decline as they move towards advanced stages. These insights can be useful in improving user engagement and overall platform performance.